

ARTICLE

Outcomes of cataract surgery in eyes with ocular melanoma treated with iodine-125 brachytherapy



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Purpose: To evaluate visual and surgical outcomes of cataract surgery in eyes with a history of iodine-125 (I^{125}) brachytherapy for ocular melanoma.

Setting: Department of Ophthalmology, David Geffen School of Medicine at UCLA and the Stein Eye Institute, Los Angeles, California, USA.

Design: Retrospective case series.

Methods: Patients with ocular melanoma treated by I^{125} brachytherapy who subsequently had cataract surgery were evaluated. The recorded data included tumor size, location, preoperative ocular comorbidities, corrected distance visual acuity (CDVA), operative complications, and brachytherapy-related maculopathy before and after surgery.

Results: Thirty-two eyes of 32 patients were included. The mean age at the time of cataract surgery was 66.1 years. The median follow-up was 53.5 months. There were no intraoperative

complications. Eighteen eyes (56.3%) had a history of preoperative radiation retinopathy, 10 involving the macula. Between 2 weeks and 4 weeks postoperatively, 22 eyes (68.8%) had an improvement in CDVA (≥ 2 lines). Seven of 10 eyes that failed to improve had radiation maculopathy. By the last follow-up examination, 13 eyes (40.6%) had improved CDVA, 9 eyes (28.1%) were worse (≥ 2 lines), and 10 eyes (31.3%) were unchanged (within ± 1 line). Of 15 eyes that lost CDVA gains achieved between 2 weeks and 4 weeks postoperatively, 9 eyes had new-onset or worsening maculopathy. Cataract surgery had no effect on local tumor control or distant metastasis.

Conclusions: Cataract surgery after I^{125} brachytherapy for ocular melanoma improved CDVA in most eyes during the immediate postoperative period. Gains were often lost with further follow-up. Progression of radiation maculopathy was primarily responsible for subsequent visual decline.

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Iodine-125 (I^{125}) brachytherapy is preferred over enucleation for the treatment of medium-sized ocular melanomas. This is one of the major outcomes of the Collaborative Ocular Melanoma Study (COMS), a randomized multicenter prospective clinical trial that found no significant difference in mortality between a group of patients treated by I^{125} brachytherapy and a group treated by enucleation.^{1,2}

Iodine-125 brachytherapy provides a favorable cosmetic outcome via preservation of the globe without decreasing lifespan and offers the benefit of vision preservation.³ Although the affected eye is saved, patients might experience a decline in visual acuity after brachytherapy because

of radiation-related complications. These include cataract development, neovascular glaucoma, radiation retinopathy, and optic neuropathy. Brachytherapy has no known systemic side effects and emits low energy photons with a favorable dose distribution.^{4,5}

Cataract surgery outcomes were the focus of this study. Cataract development is a common complication of ocular radiation.^{6,7} Several studies have reported risk factors for cataractogenesis, which include age; sex; the radiation dose to the lens; and tumor characteristics (thickness, diameter, and location).^{7–13} The COMS report number 27¹⁴ determined that cataract development and surgery occurred earlier in the oldest subgroup (≥ 65 years)

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